

2004 AUSTRALASIAN SCHOOLS

YEAR
5

SCIENCE

COMPETITION

AUSTRALIA

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

Read the instructions on the **ANSWER SHEET**
and fill in your **NAME, SCHOOL YEAR, GENDER**
and the **LANGUAGE YOU FIRST SPOKE**.

Use the information provided to choose the **BEST** answer from
the four possible options.

On your **ANSWER SHEET** blacken the oval that matches the
answer you choose for each question.
Mark only **ONE** answer for each question.

Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

Use a 2B or B pencil.
Do **NOT** use a biro or pen.
Rub out any mistakes completely.
You may use a calculator and a ruler.

THE UNIVERSITY OF
NEW SOUTH WALES



EDUCATIONAL
TESTING CENTRE

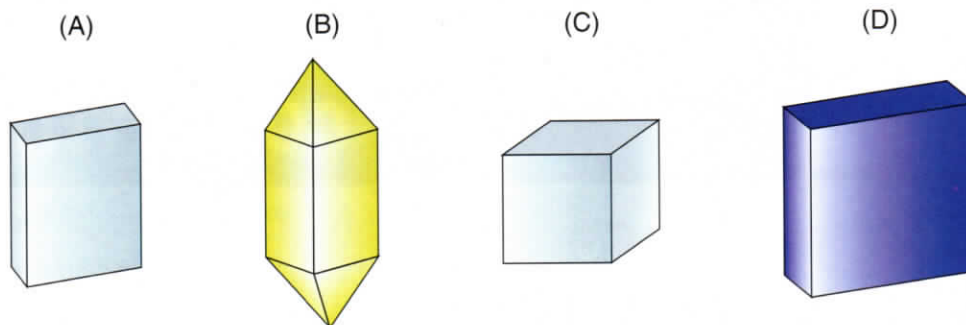
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Australian Science
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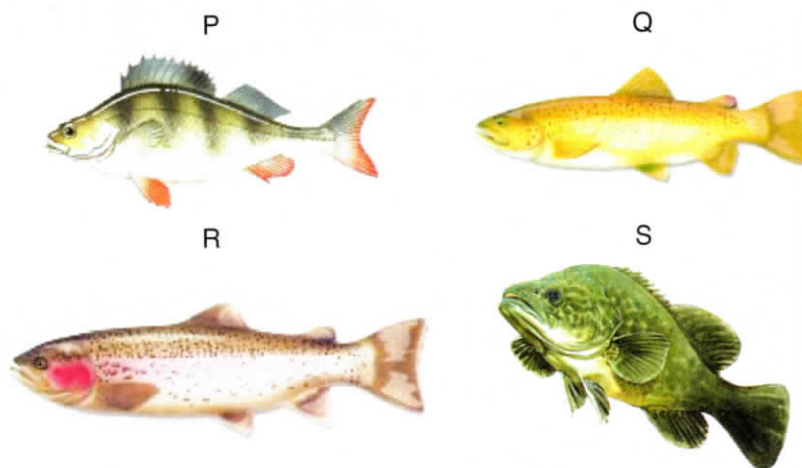
- 1 Crystals of the salt, sodium chloride, are cube shaped and colourless.

Some children drew the crystals of four salts they had examined using a magnifying glass. Which is most likely a drawing of a sodium chloride crystal?



- 2 The more closely one type of animal looks like another type of animal, the more closely related the animals are likely to be.

The drawings show four different types of fish found in a river.



Which two types of fish are likely to be most closely related?

- (A) P and R (B) P and S (C) Q and R (D) Q and S

- 3 You can see through a fat stain on brown paper.

The table gives information about the stains left on brown paper by some food.

Food	Can you see through the food stain on the brown paper?
boiled potato	no
cashew nuts	yes
full cream milk	yes
maple syrup	no

Which foods contain fat?

- (A) boiled potato and cashew nuts
(B) cashew nuts and full cream milk
(C) full cream milk and maple syrup
(D) boiled potato and maple syrup

4

The two photographs show the same whale at different times.

Photo 1 shows the whale breaching.



Photo 2 shows the whale sailing.



Which whale is shown in the photographs?

(A)



(B)



(C)

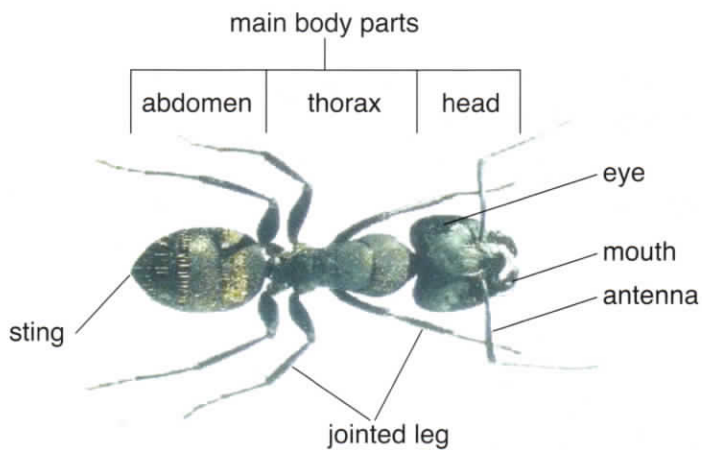


(D)



5

The photograph shows some parts of an ant.



On which part of the ant's body is the sting located?

- (A) abdomen
- (B) head
- (C) mouth
- (D) thorax

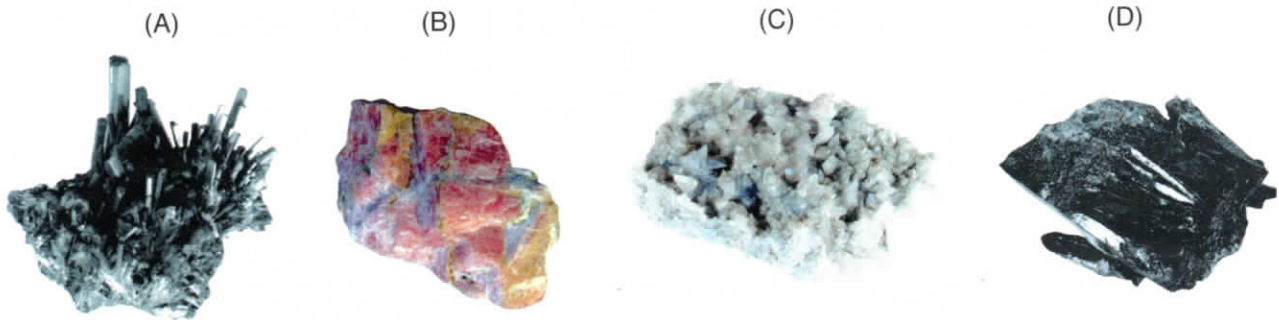
Ant image © Ant Image Database (Japan)

6

The table gives some characteristics of four minerals.

Mineral	Colour	Lustre	Crystal habits
calcite	white	glassy	small crystals of variable size
graphite	black	metallic	thin flakes
rhodochrosite	pink, yellow	glassy	round and curved faces
stibnite	steel grey	metallic	long crystals in clusters

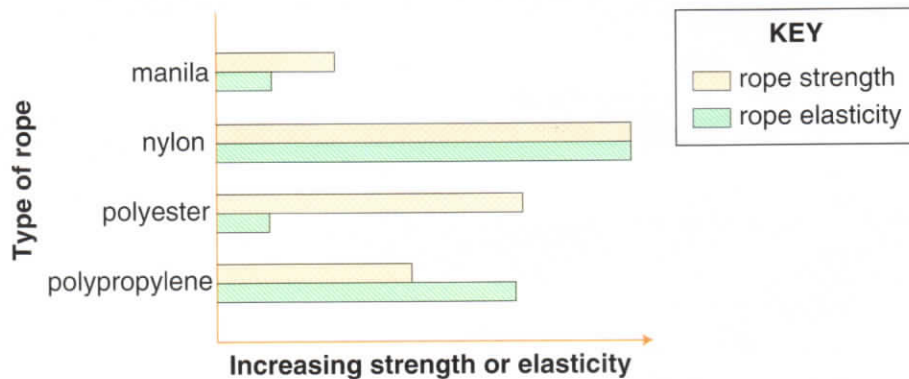
Which photograph shows a sample of stibnite?



7

Rope manufacturers define elasticity as the rope's ability to stretch over its length to absorb energy.

The graph compares the strength and elasticity of some ropes.



A tow rope needs to be strong and elastic so that it can absorb the shock as towing begins.

Which rope would be best for towing?

- (A) manila
- (B) nylon
- (C) polyester
- (D) polypropylene

8

The photo shows the actual size of Josie's pet terrapin.

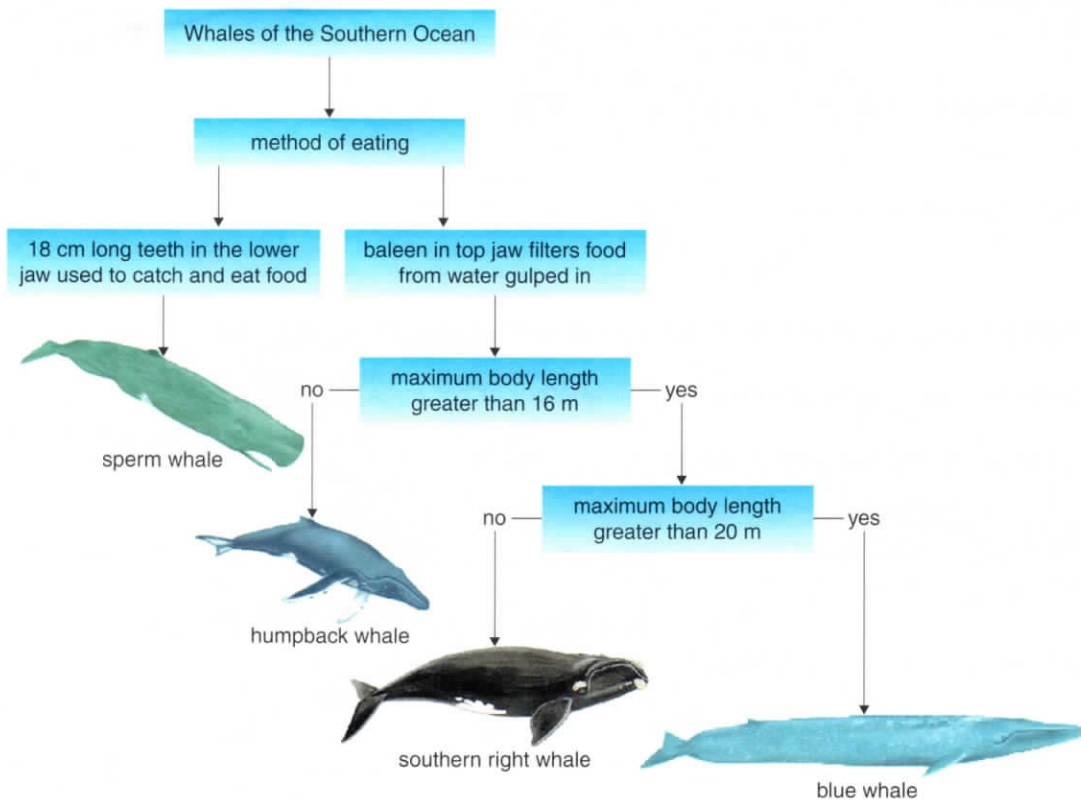
How long is the terrapin's shell?

- (A) 5 cm
- (B) 6 cm
- (C) 7 cm
- (D) 8 cm



For questions 9 and 10 use the information below.

The flowchart distinguishes between some of the whales found in the Southern Ocean.



9

Which characteristics could best describe a southern right whale?

- (A) teeth present, maximum length 18.5 m
- (B) baleen present, maximum length 13.7 m
- (C) teeth present, maximum length 13.7 m
- (D) baleen present, maximum length 18.5 m

10

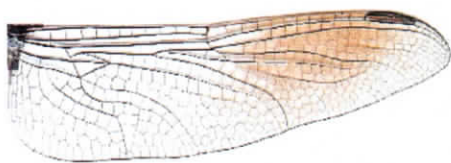
The illustrations of the four whales in the flowchart, drawn at the same scale, compare their maximum lengths.

What would be the approximate maximum length of the blue whale?

- (A) 13 m
- (B) 19 m
- (C) 21 m
- (D) 29 m

11

The photograph shows a ruler being used to measure a wing of a dragonfly.



What is the length of the wing?

(A) 3.4 mm

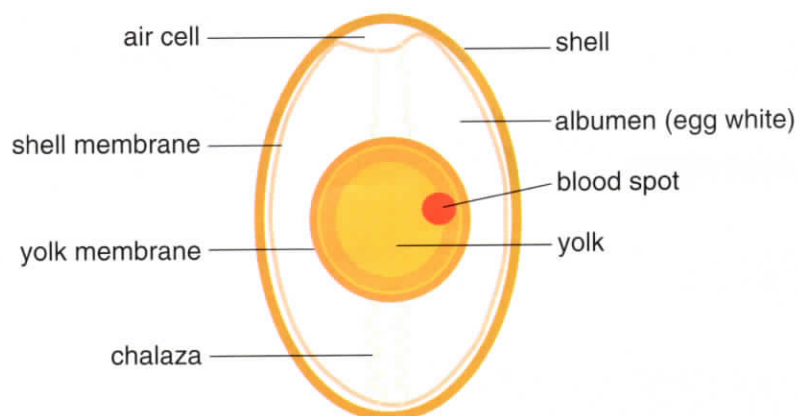
(B) 3.5 mm

(C) 34 mm

(D) 35 mm

12

The drawing shows the parts of a hen's egg.



The chalazas are spiral bands that keep the yolk in the centre of the albumen.

The table gives information about how to tell whether or not eggs are fresh.

Egg freshness	Chalazas	Size of the blood spot in the yolk (where present)	Air cell
fresh	thick	large	small
not fresh	thin	small	large

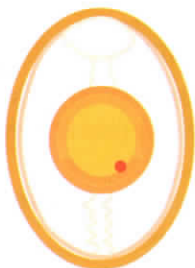
Which egg is most likely the freshest?

(A)

(B)

(C)

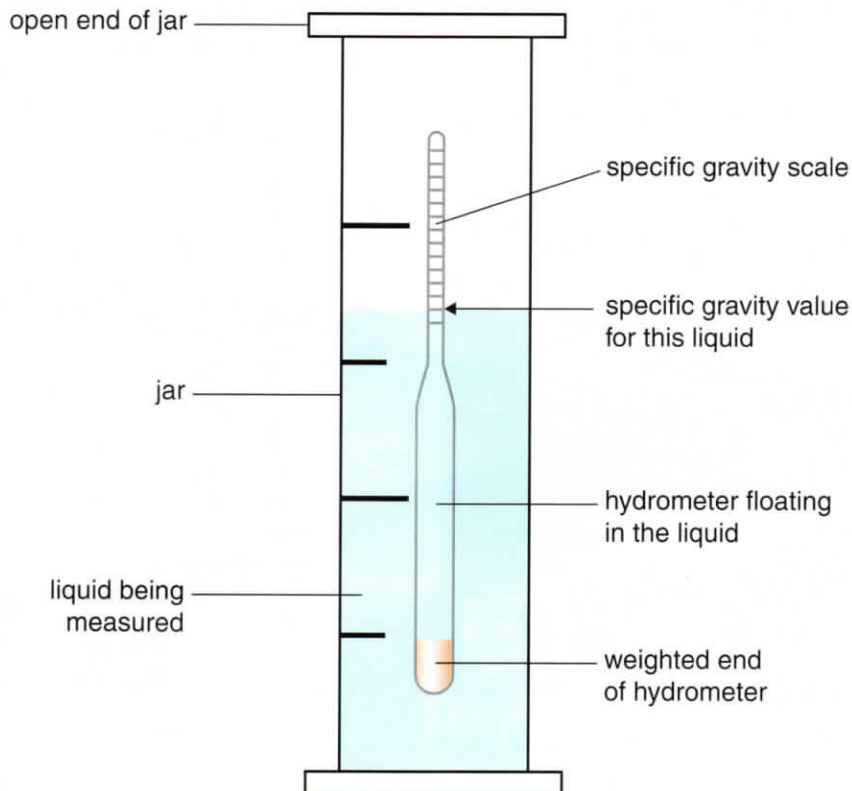
(D)



For questions 13 and 14 use the information below.

Specific gravity is measured using a hydrometer. The hydrometer compares the density of the liquid in which it is floating to the density of pure water.

The diagram shows a hydrometer being used to measure the specific gravity of a liquid.

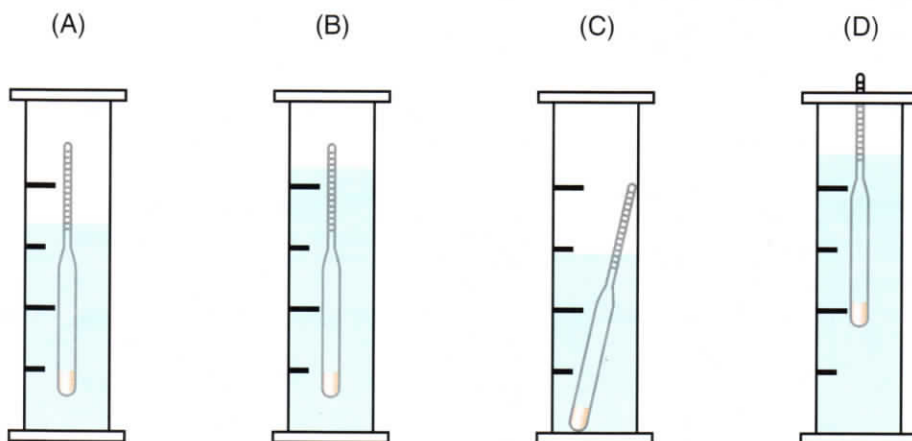


13 The specific gravity value for this liquid would be read from the scale on the

- (A) jar at the level of the liquid's surface.
- (B) jar at the level of the top of the hydrometer.
- (C) hydrometer at the level of the liquid's surface.
- (D) hydrometer at the top of the open end of the jar.

14 The diagrams show the apparatus used to measure specific gravity in four different liquids.

Which specific gravity measurement would be **incorrect**?

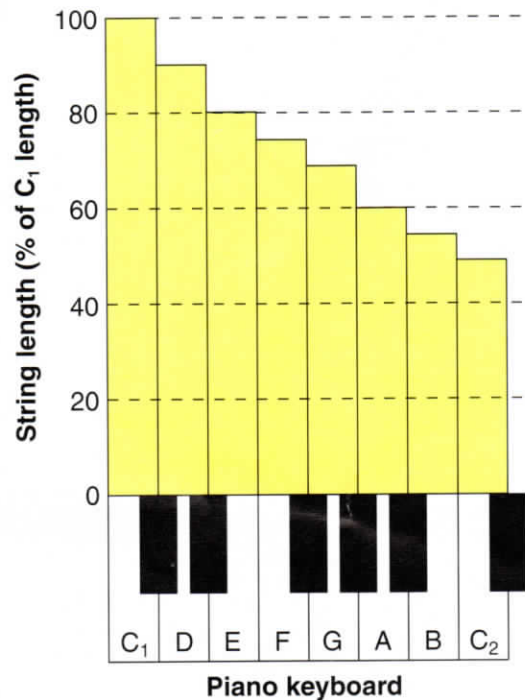
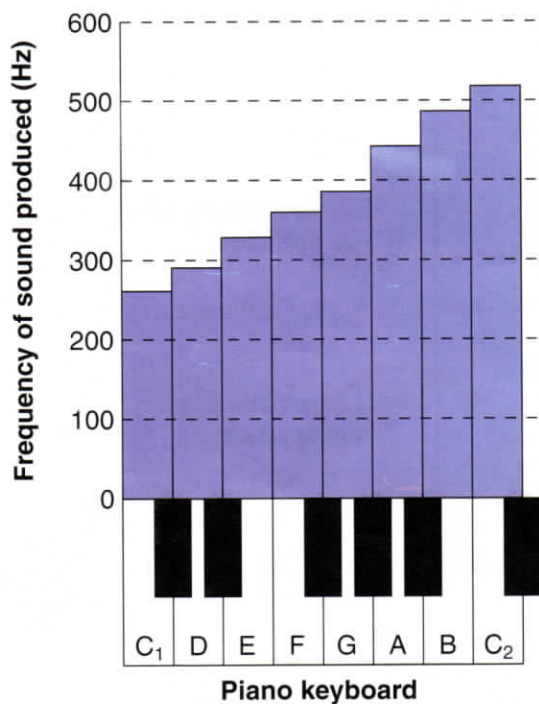


For questions 15 and 16 use the information below.

Pianos produce sound when their strings are caused to vibrate.

A string's frequency, measured in hertz (Hz), is the number of times it vibrates to and fro in a second. Different frequencies of sound are produced by varying the length of the piano's strings.

The graphs show the frequencies of the sounds produced, and lengths of the strings (compared to the length of the C_1 string) for some of the white keys on a piano keyboard.



15 What is the approximate frequency of the sound produced by the A key on the piano keyboard?

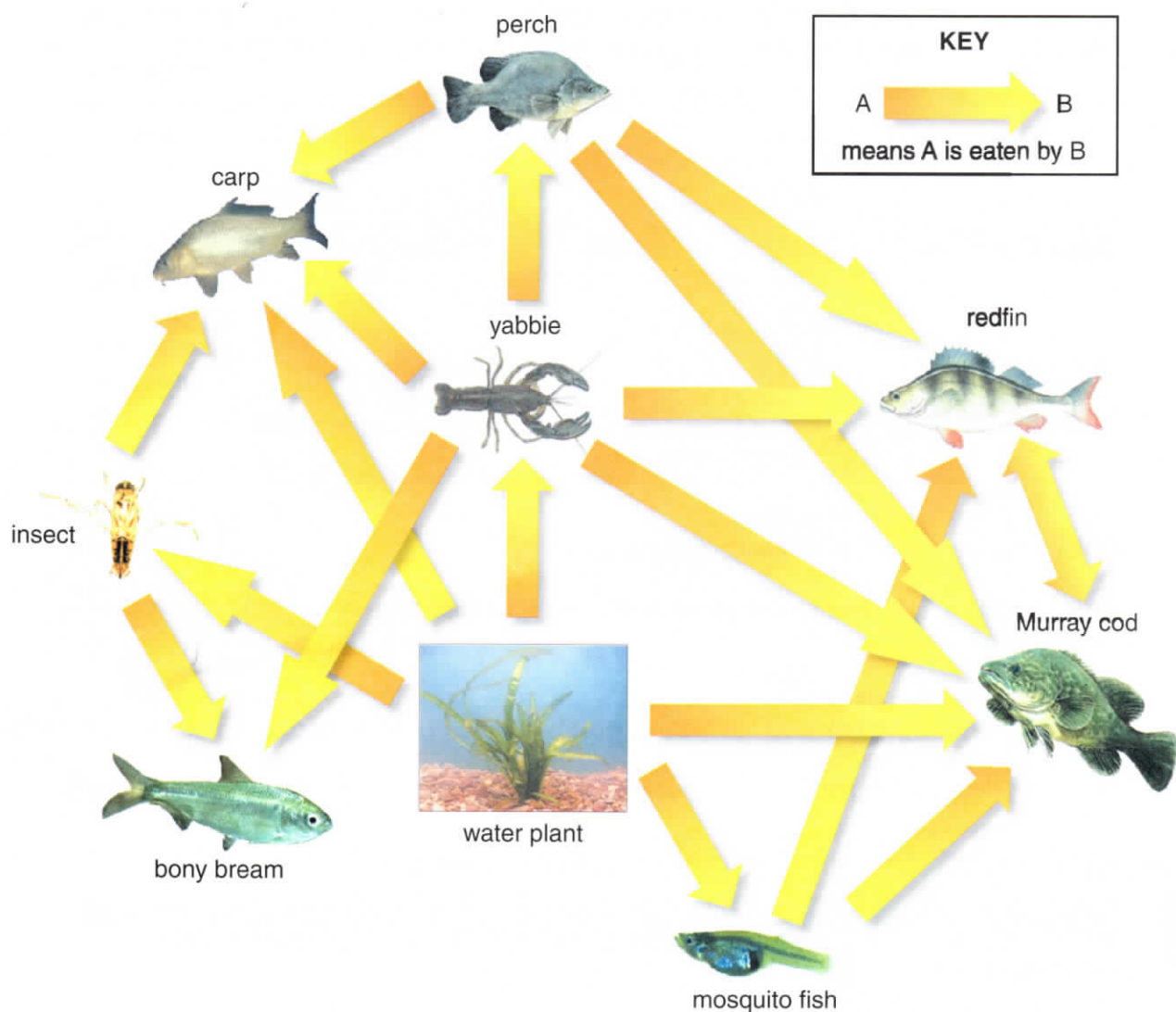
- (A) 60 Hz
- (B) 75 Hz
- (C) 370 Hz
- (D) 440 Hz

16 Compared to key C_1 , key C_2 produces a sound at

- (A) half the frequency using a string half the length.
- (B) half the frequency using a string double the length.
- (C) double the frequency using a string half the length.
- (D) double the frequency using a string double the length.

For questions 17 and 18 use the information below.

The diagram shows a food web in a freshwater river.



17 What is the perch eaten by?

- (A) yabbies only
- (B) yabbies, carp and redfin only
- (C) carp and redfin only
- (D) carp, redfin and Murray cod only

18 Introduced animals can threaten the survival of native animals by competing with them for food and by eating them.

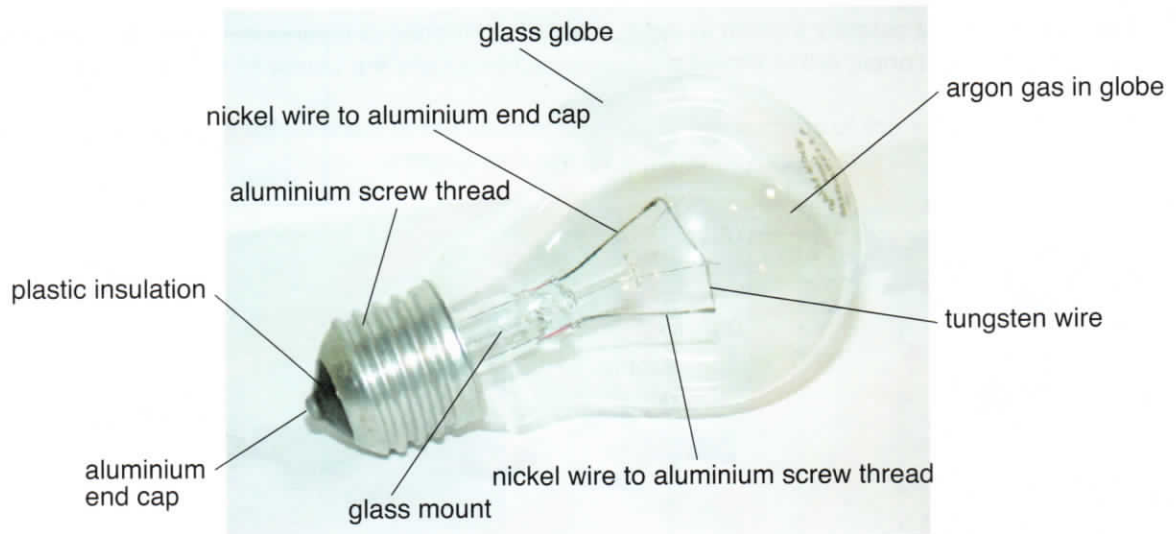
The table gives information about the animals in the food web.

Native animals	Introduced animals
bony bream, insect, Murray cod, perch, yabbie	carp, mosquito fish, redfin

Which introduced animal most threatens the Murray cod?

- (A) carp
- (B) mosquito fish
- (C) perch
- (D) redfin

- 19 The thinnest wire in an electrical circuit will resist the flow of electricity most and may become hot. If this thin wire gets hot enough it will glow, and in the presence of oxygen the wire will burn.

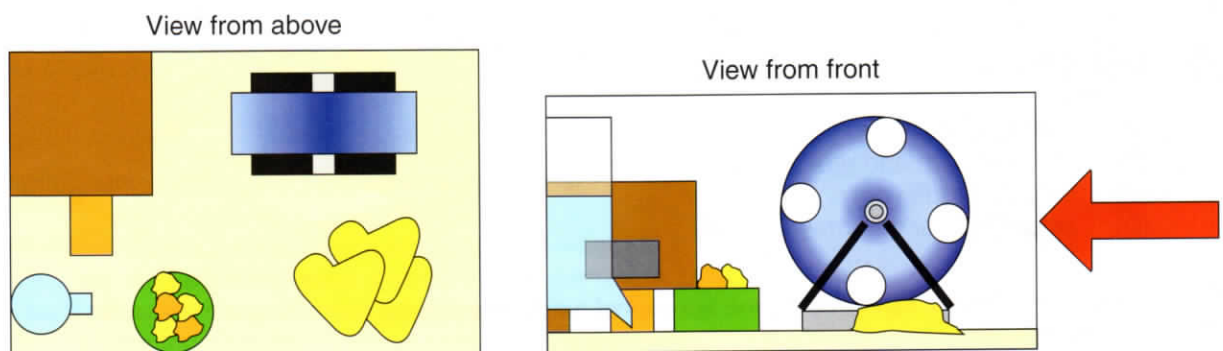


Light globes glow and give off heat but do not burn.

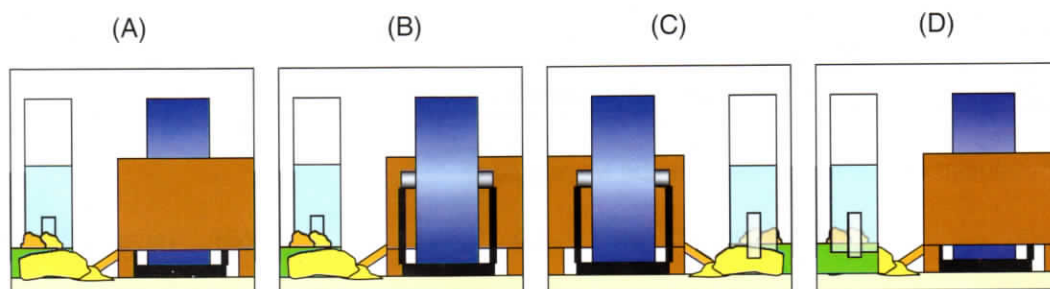
From what material is the wire that glows made, and what substance in the globe stops this wire from burning?

- (A) nickel and oxygen gas
- (B) nickel and argon gas
- (C) tungsten and argon gas
- (D) tungsten and oxygen gas

- 20 John's mouse house was contained in a clear plastic box with no lid. John drew two different views of the mouse house.



What does the mouse house look like from the side indicated by the red arrow?



For questions 21 and 22 use the information below.

Tom and Rob wanted to make their playground safer by reducing the number of accidents.

They predicted that they could reduce the number of accidents happening in future, by identifying the piece of playground equipment involved in most accidents, and then working to make it safer.

The table records the number of accidents that occurred in one week for each piece of playground equipment.

Day	Equipment			
	Rope swing	Slippery slide	Monkey bars	Adventure fort
Monday	2	0	3	2
Tuesday	1	1	2	1
Wednesday	1	0	3	2
Thursday	0	0	3	2
Friday	0	1	2	1

21 Which piece of equipment appears to be the safest?

- (A) rope swing
- (B) slippery slide
- (C) monkey bars
- (D) adventure fort

22 Tom and Rob acted to make one piece of equipment safer.

The table below records the accident numbers for the next week.

Day	Equipment			
	Rope swing	Slippery slide	Monkey bars	Adventure fort
Monday	2	1	1	3
Tuesday	0	1	2	0
Wednesday	1	0	2	2
Thursday	1	0	1	2
Friday	0	0	1	1

Tom and Rob compared the information for the first and second weeks. They concluded that their idea had been correct.

Which information did they consider in reaching this conclusion?

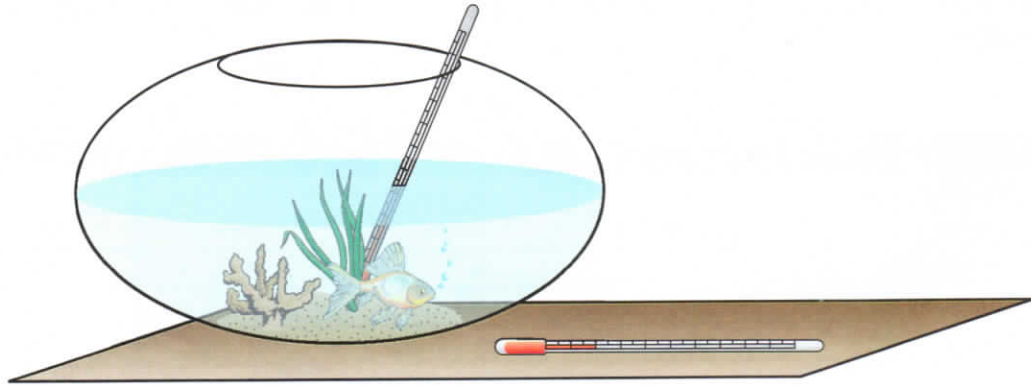
They compared the number of accidents that occurred at the

- (A) adventure fort only.
- (B) monkey bars only.
- (C) monkey bars and adventure fort only.
- (D) rope swing, slippery slide, monkey bars and adventure fort.

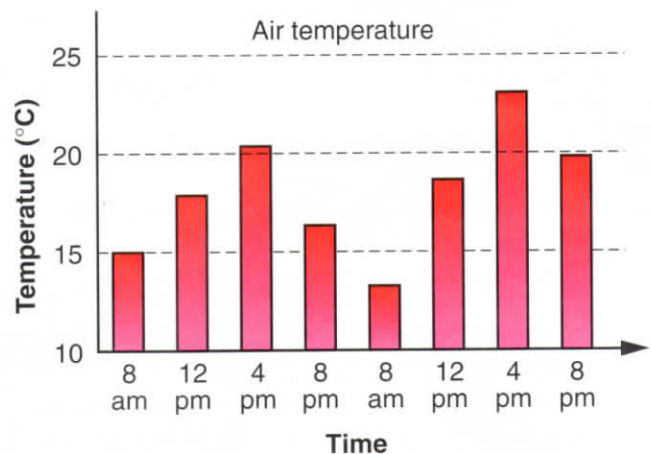
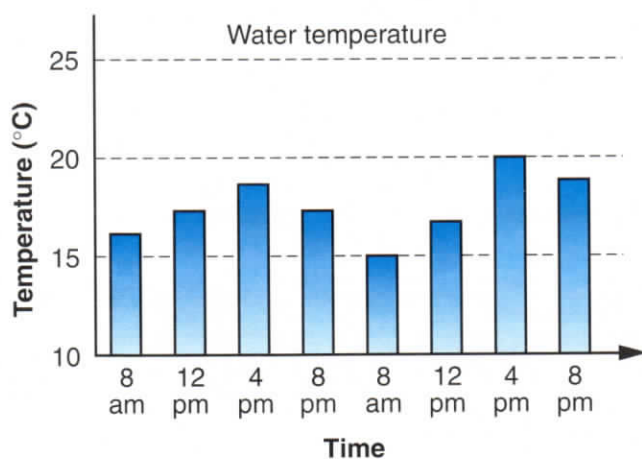
For questions 23 and 24 use the information below.

Amber predicted that the temperature in her fish bowl varied less than the temperature of the air nearby.

She set up her experiment as shown to test her idea.



Amber measured the temperature in each location for the next two days. The two graphs show her results.



- 23 What is a correct statement that Amber could make about her results?
- (A) Each day the air temperature was highest at 4 pm.
 - (B) Sunlight heated the water more quickly than the air.
 - (C) The air temperature at 8 am on the second day was 15 °C.
 - (D) The air and water temperatures were the same at 8 pm on the first day.

- 24 Was Amber's prediction correct? What is the **difference** between the temperature ranges in the water and in the air over the two days?
- (A) Yes. 5 °C
 - (B) Yes. 10 °C
 - (C) No. 10 °C
 - (D) No. 5 °C

For questions 25 and 26 use the information below.

An animal's dental formula gives the number and type of permanent teeth for the top jaw and bottom jaw on one side of its mouth. For each tooth type, the first number indicates teeth numbers in the top jaw, and the second number indicates teeth numbers in the bottom jaw.

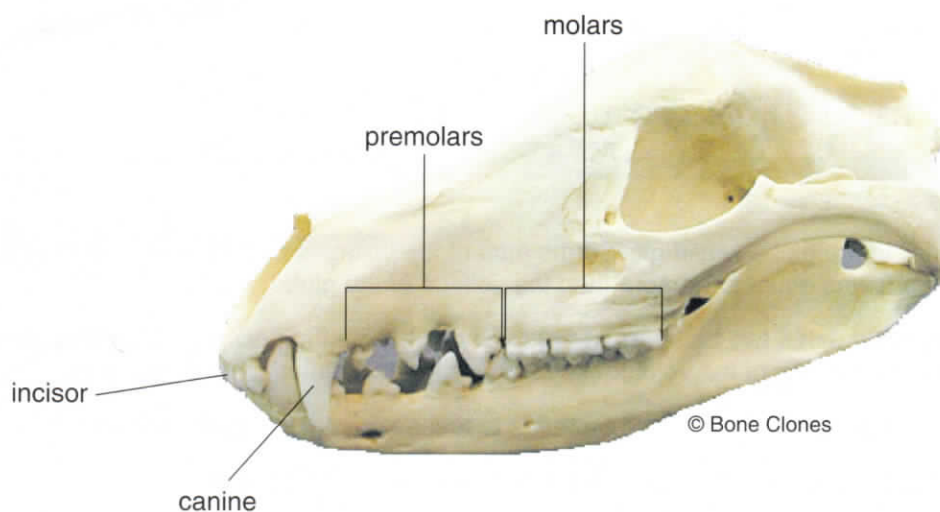
Adult humans usually have 32 permanent teeth.

The dental formula for an adult human is $I - 2/2 \quad C - 1/1 \quad P - 2/2 \quad M - 3/3$

The table gives the dental formula for some adult animals.

		Animal				
Number of teeth	Tooth type	Human	W	X	Y	Z
	incisor (I)	2/2	0/4	3/3	3/3	3/3
	canine (C)	1/1	0/0	1/1	1/1	1/1
	premolar (P)	2/2	3/3	4/4	3/4	4/4
	molar (M)	3/3	3/3	2/3	3/3	3/3

The photograph shows the skull of an adult thylacine. No teeth are missing from the skull.



- 25 One of the animals (W, X, Y, Z) is a thylacine. Which one is it?

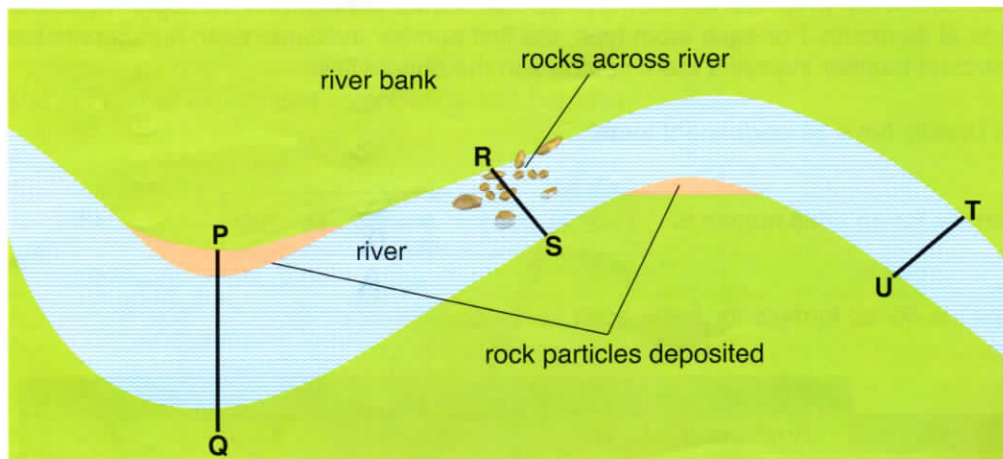
(A) W
(B) X
(C) Y
(D) Z

- 26 How many teeth does animal Z have?

(A) 21
(B) 22
(C) 42
(D) 44

27

When a river is **not** rising or falling, the same volume of water flows past any given location in the same time. The map shows a part of a river.



The cross-section diagrams show the river bottom at locations PQ, RS and TU. The cross-section diagrams are at a different scale to the map.



List the cross-section locations in the order that shows the location with the fastest water flow to the location with the slowest water flow.

- (A) PQ, RS, TU (B) PQ, TU, RS (C) RS, PQ, TU (D) RS, TU, PQ

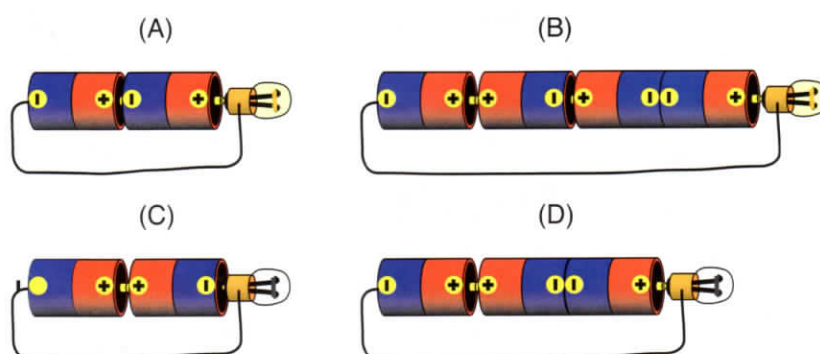
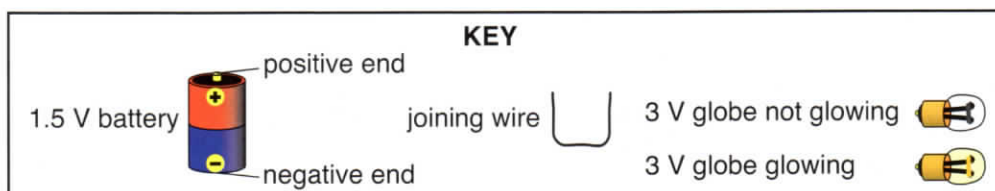
28

Mindy knew that a globe would glow, if the sum of the voltages of the batteries (joined positive end to negative end) was equal to or more than the voltage of the globe.

Mindy predicted that if two batteries were joined positive end to positive end (or negative end to negative end) the sum of the voltage produced by the two batteries would be zero.

She used 3 volt globes and 1.5 volt batteries to test her idea. Her results are shown below.

If Mindy predicted correctly, which result would she **not** expect to see?



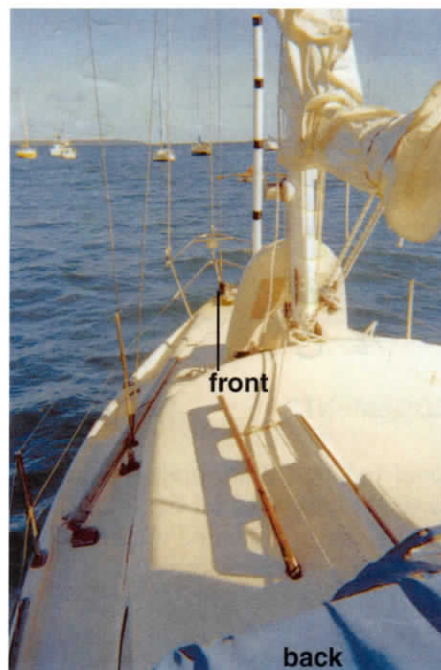
29

The sun rises in the east and sets in the west.

The photograph shows the view along the deck of a boat in the late afternoon. The view is from the back to the front of the boat. The boat is moving in the direction the front is pointing.

In which direction is the boat moving?

- (A) north
- (B) south
- (C) east
- (D) west

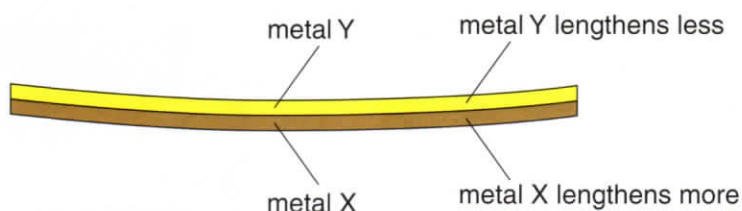


Photograph courtesy of G. Rickard

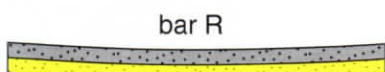
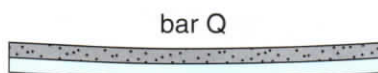
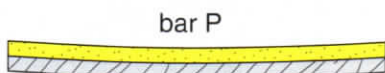
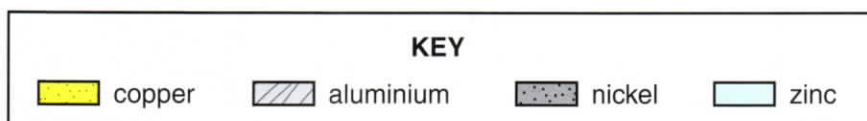
30

Jim knew that metals lengthen when heated, and that different metals lengthen at different rates.

Jim thought that if he joined two metal strips of the same size together into a bar and heated them, they would bend as shown in the diagram. He could then compare the rates at which the metals lengthened.



Jim heated four of these bars for two minutes. His results are shown below.



Jim concluded that his idea had been correct, and that the order listing the metals from that which lengthened most, to that which lengthened least, was

- (A) aluminium, copper, nickel and zinc.
- (B) aluminium, nickel, zinc and copper.
- (C) zinc, aluminium, copper and nickel.
- (D) zinc, nickel, copper and aluminium.

Year 5 - ICAS SCIENCE

2004 - Paper B

- | | | | |
|------|-------|-------|-------|
| 1. C | 9. D | 17. D | 25. C |
| 2. C | 10. D | 18. D | 26. D |
| 3. B | 11. C | 19. C | 27. D |
| 4. B | 12. A | 20. B | 28. B |
| 5. A | 13. C | 21. B | 29. B |
| 6. A | 14. C | 22. D | 30. C |
| 7. B | 15. D | 23. A | |
| 8. B | 16. C | 24. A | |

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